Lane Lords — Technical Sheet

Tech used

Single self-contained HTML file, no external assets or libraries. HTML5 Canvas 2D renderer with devicePixelRatio scaling (capped at 2x), additive-blend glow, particle system (capped at 900), screenshake and floating damage numbers.

Web Audio API: all sound effects synthesized (oscillators + filtered noise) and a procedural synthwave soundtrack whose tempo and density rise with combat tension near enemy heroes. Audio starts on first tap (iOS-safe) and suspends when the tab is hidden.

Pointer Events for multitouch: floating virtual joystick plus independent ability buttons; safe-area insets for notched phones; localStorage for win count.

Game systems

3-lane map (2400x2400) with 12 towers, 2 Cores, river and decorations. 3v3 heroes with AI allies and rivals (lane holding, retreat-and-heal, chase/trade decisions, ability use, auto shopping).

Minion waves with lane pathing, melee/ranged/big types, separation steering and per-wave scaling plus Overdrive after 8 minutes.

Tower aggro rules, backdoor protection (35% hero damage without minions), XP/level curve to 12, gold economy (passive, last hits, kills, towers), 6-item shop, recall, respawn timers, 3 difficulties (rival stat scaling and AI aggression).

Audit 1 — Balance (simulated play)

Ran full bot-vs-bot matches (all 6 heroes on AI, every difficulty) through a headless copy of the game.

First version: towers and Cores were so tough that matches stalled for 25+ minutes with only 0-2 towers down. Fixes: tower HP cut from 1600-2100 to 800-1100, Core HP 4000 to 2400, Core damage 110 to 70, minions deal double damage to structures, towers deal 45% less to minions, waves every 25s instead of 30s.

Added Overdrive (minions scale faster after 8 minutes) and Sudden Death (structures take 4x damage after 10 minutes) so no match can stall.

Result on the final build: every simulated match ended, typically in 8-22 minutes (bot heroes defend hard; a human pushing with minions finishes faster), and wins were split between the two teams with no side advantage.

Audit 2 — Bugs

Crash fixed: ability area rings were drawn before the ring list existed, which would have crashed the first ultimate cast.

Recall progress ring could draw past a full circle; now clamped.

Keyboard W conflicted between move-up and the W ability; W is now ability-only (arrows move).

Audio is optional: the game runs silently on browsers without Web Audio, starts sound on the first tap (iPhone-safe) and pauses when the tab is hidden.

Audit 3 — Player experience

Ability buttons show a dark countdown overlay while on cooldown; abilities auto-aim so one thumb is enough, or fire in your joystick direction.

Shop button dims outside base and explains why when tapped; items you can't afford are greyed out.

On-screen toasts for level-ups, towers lost/destroyed, hero kills, Overdrive and Sudden Death, plus a respawn countdown, so nothing happens without explanation.